Replication of Study 1 by Estes, Z., Verges, M., & Barsalou, L.W. (Psychological Science, 2008).

Head up, foot down. Object words orient attention to the objects' typical location. *Psychological Science, 19*(2), 93–97.

Replication Authors

Frank Renkewitz

University of Erfurt

frank.renkewitz@uni-erfurt.de

Stephanie Müller

University of Erfurt

stephanie.mueller.01@uni-erfurt.de

Introduction

Some words denote objects that typically occur at specific locations. For instance, branches are typically overhead, whereas roots are underfoot. Thus, such words have a particular "spatial connotation". Estes, Verges and Barsalou tested in three experiments whether such object words orient attention to the corresponding locations. Their specific hypothesis was that such an attentional effect of object words should *interfere* with the visual perception of a target in the typical location of the object. This hypothesis was based on the assumption that object words evoke a perceptual simulation. A perceptual simulation of an object should interfere with the perception of another visual stimulus that requires the same neural mechanisms. Therefore, "(…) if an object word activates a perceptual simulation in the object's typical location, then that word should hinder perception of a visual target in that location. Essentially, because the perceptual mechanisms required for target identification are engaged in the simulation of the denoted object at its typical location, identification of a target at that location should be delayed". (p. 94).

We will replicate Experiment 1. "In this experiment, object words associated with an upper or lower location served as cues. To ensure that the object words unambiguously denoted an upper or lower location, we presented a context word before each object word. Thus, on each trial, a context word (e.g., cowboy) was followed by an upper location (e.g., hat) or lower-location (e.g., boot) cue. Context and cue words were both presented centrally. A target letter (X or O) then appeared at the top or bottom of the display, and participants identified the target as quickly as possible. (p. 95)

Methods

Power Analysis

Estes, Verges and Barsalou compare targets occurring at typical (e.g., bottom targets after a lower-location cue) and at atypical (e.g., bottom targets after an upper-location cue) locations. They present within-subjects analyses of variance of response times and error-rate data. For response times, the reported effect size is $\eta_p^2 = .70$; for error-rates it is $\eta_p^2 = .35$. Power analysis (G*Power (Version 3.1): *ANOVA: Repeated measures, within factors*) indicates that for the effect in error rates a power of 80%, 90%, and 95% is achieved with sample sizes of 6, 8, and 9, respectively. For the effect in response times, 4 participants would yield a power of 97% (assuming a correlation of $r = .5$ between repeated measures in all power calculations).

Planned Sample

The planned sample size for the replication study is $N = 20$. This ensures a power of more than 95% for both effects. Participants will be undergraduates at the University of Erfurt, mostly with a major in psychology. They will receive course credit for participation.

Materials

The experimental stimuli were translated into German by the authors (see table 1 in the Appendix). Materials were described by Estes, Verges, and Barsalou as follows:

> The experimental stimuli were 30 context words, each paired with one upper-location and one lower-location cue word (i.e., 60 spatial cues). We also included 60 filler stimuli consisting of 30 context words, each paired with 2 nonspatial cue words (e.g., chocolate powder, chocolate shavings). (…) Ten practice trials preceded the 120 experimental trials. (p. 94)

Procedure

The experimental procedure of the replication study will correspond exactly to the original study. Instructions were translated into German (see table 2 in the appendix for detailed instructions and their translation).

> Participants initiated each trial by pressing the space bar, which triggered a central fixation cross that appeared for 250 ms. The context word then appeared centrally for 500 ms, replaced immediately by the cue word for 250 ms. After a 50-ms delay, a target letter subtending approximately 1° of visual angle appeared at the top or bottom of the screen. The "top" and "bottom" locations were centered horizontally approximately 8° vertically from the center of the display. Participants were instructed to identify each target letter as quickly and accurately as possible by pressing the appropriate key. Location cue (upper, lower), target location (top, bottom), and target letter (*X*, *O*) were fully crossed and balanced, such that each target letter was equiprobable at each target location, and each target location was equiprobable within each cue condition. (p. 94)

Analysis Plan

We will analyze response times and error rates using within-subjects analyses of variance. Incorrect trials and response times greater than 1,000 ms will be excluded from the analysis of response times. Following the analysis plan of the original article, we will additionally perform analyses of variance across items (F_2) with both kinds of data.

Differences from Original Study

The only difference from the original study is the translation of instructions and experimental stimuli into German.

(Post Data Collection) Methods Addendum

Actual Sample

The actual sample size of the study were 22 participants (instead of 18 in the original study), none were excluded from the analysis.

Differences from pre-data collection methods plan

None

Results

Data preparation

Target locations were coded as typical or atypical. In accordance with the original study, incorrect trials (3.4% of trials) and response times greater than 1,000 ms (1.0% of trials) were excluded from the analysis of response times.

Confirmatory analysis

We found no differences in response times or error rates between targets that appeared in typical or atypical locations of the objects denoted by the preceding word (see also Figure 1). In the response time data, the effect of location was non-significant, $F_1(1, 21) = 0.001$, $p = .97$, and $F_2(1, 58) = 0.12$, $p = .73$, and close to zero (-0.23 ms, $\eta_p^2 < .001$ (F_1) or $\eta^2 = .002$ (F_2)). In the error rate data, this effect was also non-significant, $F_1(1, 21) = 1.45$, $p = .24$, and $F_2(1, 58) = 1.40$, $p = .24$, and again small (1.36%, $\eta_p^2 = .065$ (F_1) or $\eta^2 = .024$ (F_2)).

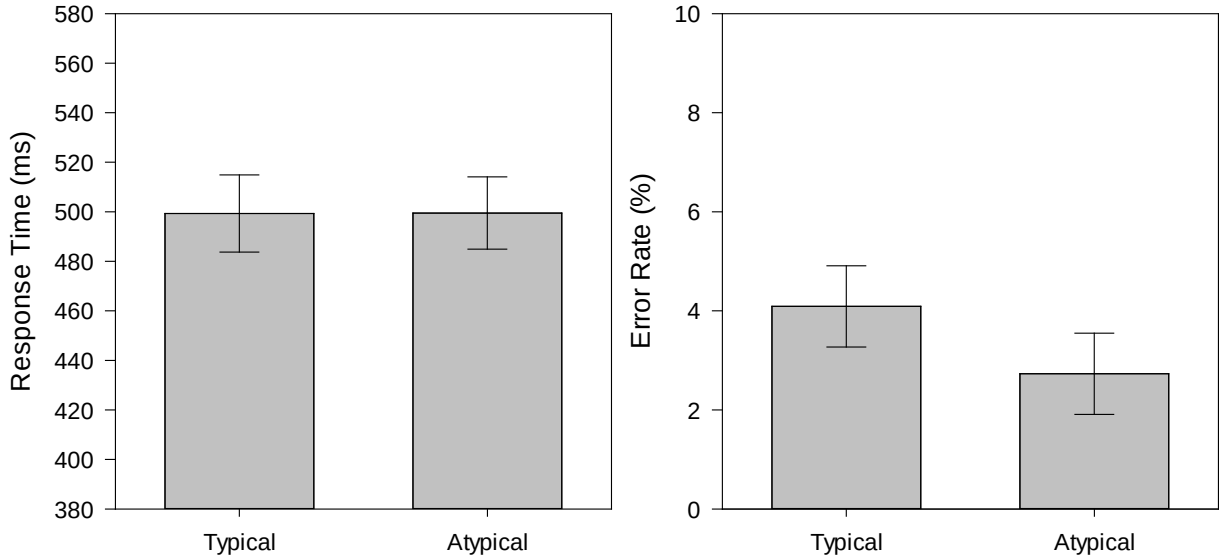


Figure 1: *Mean response times (left) and error rates (right) as a function of target location (typical, atypical). Error bars show standard errors.*

Exploratory analyses
We did not conduct any follow up analyses.

Discussion

Summary of Replication Attempt

We could not replicate any of the two effects reported in Study 1 by Estes, Verges and Barsalou (2008). In our replication attempt, the effect of target location on response time and error rate was not only non-significant but almost zero. Thus, there was no evidence for the assumption that the cue word evoked a perceptual simulation in the object's typical location that was hindering perception of a target letter at that location.

Commentary
A possible explanation for our failure to replicate the original results of Estes, Verges and Barsalou (2008) may be the translation of the stimulus material into German. In our judgment, many of the German cue words do not have a strong association with a specific (upper or lower) location (e.g. Halm, Baum, Stiel, Blütenblatt, Schnürsenkel; Räder, Radio, Milch, Kirchenbank, Schublade). At the same time, it seems doubtful to us whether these associations are stronger for the original English cue words (handle, tree, stalk, petal, laces – all upper location; wheels, radio, milk, pew, drawer – all lower location). This holds especially when these cue words are presented without a disambiguating context word as was successfully done in Experiment 3 of the original article. Finally, it seems noteworthy us, that the original instructions included a debriefing section that described the opposite of the hypothesis given in the final article (i.e., in the debriefing section cue words to typical locations were described as *facilitating* perception of target letters; see Table 2 in the appendix).

Appendix

Table 1
Experimental stimuli and their German translation

Test Stimuli

	Context		Upper-location cue		Lower-location cue	
1	bear	Bär	Snout	Schnauze	claw	Klaue
2	bedroom	Schlafzimmer	curtains	Vorhänge	carpet	Teppich
3	bicycle	Fahrrad	seat	Sitz	tire	Reifen
4	bird	Vogel	eye	Auge	talon	Kralle
5	bottle	Flasche	cap	Deckel	coaster	Untersetzer
6	broom	Besen	handle	Stiel	bristles	Borsten
7	car	Auto	windows	Fenster	wheels	Räder
8	church	Kirche	steeple	Kirchturm	pew	Kirchenbank
9	clock	Uhr	tower	Turm	radio	Radio
10	coconut	Kokosnuss	tree	Baum	milk	Milch
11	corn	Mais	stalk	Halm	seed	Samen
12	cowboy	Cowboy	hat	Hut	boot	Stiefel
13	desk	Schreibtisch	lamp	Lampe	drawer	Schublade
14	dog	Hund	collar	Halsband	paw	Pfote
15	elephant	Elefant	ear	Ohr	foot	Fuß
16	floor	Boden	lamp	Lampe	rug	Läufer
17	flower	Blume	petal	Blütenblatt	stem	Stiel
18	horse	Pferd	mane	Mähne	hooves	Hufen
19	hotel	Hotel	roof	Dach	lobby	Lobby
20	house	Haus	attic	Obergeschoss	cellar	Keller
21	mountain	Gebirge	peak	Gipfel	base	Tal
22	mushroom	Pilz	cap	Hut	stalk	Stängel
23	ocean	Meeres	surface	Spiegel	floor	Grund
24	rain	Regen	cloud	Wolke	puddle	Pfütze
25	shoe	Schuh	laces	Schnürsenkel	soles	Sohlen
26	shower	Dusche	spout	Duschkopf	drain	Abfluß

27	street	Straße	light	Licht	corner	Ecke
28	table	Tisch	cloth	Tuch	leg	Bein
29	tree	Baum	leaves	Blätter	roots	Wurzeln
30	trouser	Hose	buckle	Schnalle	hem	Saum

Filler Stimuli

Context		**Stimulus 1**		**Stimulus 2**	
cable	Kabel	brake	Bremse	pulley	Rolle
fruit	Frucht	food	Nahrung	slice	Stück
honey	Honig	soup	Suppe	muffin	Muffin
milk	Milch	virus	Virus	bacterium	Bakterie
moth	Motte	shape	Form	wing	Flügel
nose	Nase	sound	Geräusch	itch	Juckreiz
pinie	Pinie	oil	Öl	bark	Rinde
student	Student	vote	Wahl	accusation	Anschuldigung
sugar	Zucker	alcohol	Alkohol	extract	Extrakt
water	Wasser	bird	Vogel	algae	Algen
wood	Holz	stove	Ofen	freshener	Raumspray
coal	Kohle	town	Stadt	country	Land
cactus	Kaktus	bee	Biene	chipmunk	Streifenhörnchen
coffee	Kaffee	tension	Anspannung	stain	Fleck
chocolate	Schokolade	shavings	Raspeln	powder	Pulver
birth	Geburt	rash	Ausschlag	cramp	Zange
cane	Schilfrohr	sugar	Zucker	juice	Saft
college	Hochschule	lake	See	property	Besitz
family	Familie	cow	Kuh	garden	Garten
grain	Getreide	law	Gesetz	controversy	Streit
tax	Steuern	pressure	Druck	wealth	Wohlstand
oil	Öl	treatment	Behandlung	moisturizer	Creme
plastic	Kunststoff	report	Bericht	shortage	Mangel
steam	Dampf	lock	Sperre	turbine	Turbine

winter	Winter	book	Buch	show	Vorstellung
air	Luft	power	Kraft	current	Strom
olive	Olive	season	Jahreszeit	fertilizer	Dünger
summer	Sommer	money	Geld	clothes	Kleidung
urban	Stadt	riot	Aufruhr	tower	Turm
job	Beruf	language	Sprache	competition	Wettbewerb

Practice Stimuli

Context		**Stimulus 1**		**Stimulus 2**	
floral	Blumen	wreath	Kranz	perfume	Duft
historical	Geschichte	members	Mitglieder	elder	Ältere
marital	Ehe	instincts	Instinkte	preparation	Vorbereitung
national	National	resources	Rohstoffe	police	Polizei
paper	Papier	equipment	Ausrüstung	staples	Heftklammer

Table 2
Original instructions and their German translation.

	Original Instructions	**German Instructions**
Before practice	Thank you for participating in this experiment, which concerns language comprehension. The experiment typically takes about 10 minutes to complete. Each trial of the experiment has two parts. In the first part, you will simply read a pair of words. You won't need to respond to this word pair, just consider what it might mean. Then, after about one second, the letter 'X' or 'O' will appear at the top or bottom of the screen. For these, your task is to type the letter ('X' or 'O') as quickly as you can. Please try to respond as quickly as you can, without making errors. The computer will be timing your responses. So you get an idea of what the experiment will be like, you'll do a few practice trials before the real experiment. When you're ready to begin the practice trials, press the spacebar.	Vielen Dank für die Teilnahme an diesem Experiment! Diese Studie befasst sich mit Prozessen des Sprachverständnisses. Das Experiment dauert in der Regel in etwa 10 Minuten. Es untergliedert sich in mehrere Durchgänge. Jeder Durchgang innerhalb des Experiments besteht aus zwei Teilen. Im ersten Teil wird Ihnen ein Wortpaar gezeigt. Sie brauchen nicht auf dieses Wortpaar zu reagieren – überlegen Sie sich nur, was es bedeuten könnte. Nach etwa einer Sekunde wird der Buchstabe ‘X’ oder ‘O’ am oberen oder unteren Rand des Bildschirms erscheinen. Nun ist es Ihre Aufgabe, diesen Buchstaben (‘X’ oder ‘O’) so schnell wie möglich einzutippen. Bitte versuchen Sie, so schnell wie möglich zu antworten, ohne dabei Fehler zu machen. Der Computer wird Ihre Antwortzeit messen. Damit Sie sich ein Bild vom Ablauf des Experiments machen können, werden Sie vor Beginn des eigentlichen Experiments ein paar Übungsdurchgänge bearbeiten. Wenn Sie bereit sind, mit diesen Übungsdurchgängen zu beginnen, drücken Sie die Leertaste.
After practice	You've finished the practice trials. The real experimental trials will be just like the ones you just did. Just remember to press the 'X' or 'O' key as fast as you can upon reading the word pairs. Whenever you're ready to start the real experiment, press the spacebar, and it will begin immediately.	Sie haben nun die Übungsdurchgänge beendet. Das tatsächliche Experiment wird genauso ablaufen wie die Übungsdurchgänge, die sie gerade abgeschlossen haben. Denken Sie lediglich daran, die ‘X’ oder ‘O’-Taste nach dem Lesen der Wortpaare so schnell wie möglich zu drücken. Sobald Sie bereit sind, mit dem eigentlichen Experiment zu beginnen, drücken Sie die Leertaste und es wird sofort losgehen.
Debriefing	Thank you for participating in this experiment, which was conducted by Michelle Verges, under the direction of Dr. Zachary Estes. In the experiment, you read pairs of words, and then pressed the 'O' or the 'X' as quickly as possible. On some trials, the word pair (e.g., floor lamp) had a meaning that indicated the object was above or below (e.g., floor rug). The question of this experiment is whether the up or down relation quickens or slows the reaction to respond to a stimulus that is either in an up or down location. If so, then people should respond to the 'X' or 'O' letter at the top of the screen SLOWER when they have read FLOOR RUG in comparison to FLOOR LAMP. Likewise, people should respond to the 'X' or 'O' letter at the bottom of the screen FASTER when they have read HORSE HOOVES in comparison to HORSE MANE We again thank you for your participation. If you have any further questions about this experiment, please feel free to contact Michelle Verges at mverges@uga.edu.	Vielen Dank für die Teilnahme an diesem Experiment! Diese Studie wurde von Dr. Frank Renkewitz und Dr. Stephanie Müller durchgeführt. Im Experiment haben Sie Wortpaare gelesen und anschließend so schnell wie mögliche die Tasten ‘O’ oder ‘X’ gedrückt. In manchen Durchgängen hatte das Wortpaar (z.B. Boden Lampe) eine Bedeutung, die nahelegt, dass sich das zweite Objekt eher oben oder eher unten (z.B. Boden Teppich) befindet. Die Fragestellung des Experiments war nun, ob diese Verhältnisse zwischen Oben und Unten die Reaktionen auf einen Stimulus, der entweder oben oder unten erscheint, verlangsamen oder beschleunigen. Wenn dies der Fall ist, dann sollten Menschen auf den Buchstaben ‘X’ oder ‘O’ am oberen Rand des Bildschirms LANGSAMER reagieren, wenn Sie das Wortpaar BODEN TEPPICH gelesen haben (im Vergleich zum Wortpaar BODEN LAMPE). Ebenso sollten Menschen auf den Buchstaben ‘X’ oder ‘O’ am unteren Rand des Bildschirms SCHNELLER reagieren, wenn Sie das Wortpaar PFERD HUFE gelesen haben (im Vergleich zum Wortpaar PFERD MÄHNE). Wir möchten uns nochmals für Ihre Teilnahme bedanken. Wenn Sie weitere Fragen zu diesem Experiment haben, wenden Sie sich bitte an Frank Renkewitz (frank.renkewitz@uni-erfurt.de) oder Stephanie Müller (stephanie.mueller.01@uni-erfurt).